Financial Management

By:
Dr. Vaishali Pagaria

vaishali.pagaria@wilp.bits-pilani.ac.in

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2.3: Risk Analysis in Capital Budgeting

Risk Analysis in Capital Budgeting

- Risk is inherent in almost every business decision
- Capital budgeting decisions involve costs and benefits extending over a long period of time during which many things can change in unanticipated way.
- R & D projects are more riskier than the expansion project and the latter tends to be more riskier than a replacement project.
- Therefore, risk need to be evaluated explicitly in capital budgeting decision.
- This is called **Risk Analysis in Capital Budgeting**.

Risk Analysis Techniques

1. Sensitivity Analysis
2. Scenario Analysis
3. Break-even Analysis
4. Decision Tree Analysis

Sensitivity Analysis

- Also known as ‘what if’ analysis
- It is a way of analysing change in the project’s NPV or IRR for a given change in one of the variables.
- It indicates how sensitive a project’s NPV or IRR is to change in particular variable.
- The more sensitive the NPV or IRR, the more critical is the variable
- Eg. what will happen to the viability of the project when some variable like sales or investment deviates from its expected value?
- Suppose Bombay Textile Mills is considering setting up a new flour mill near Bangalore. Based on company’s previous experience, the project staff of the company has developed the figures shown in the table (assuming salvage value to be nil). The cost of capital is 12 per cent

Steps for Sensitivity Analysis

1. Identification of all those variables which have an influence on the project's NPV or IRR
2. Definition of the underlying (mathematical) relationship between the variables.
3. Analysis of the impact of the change in each of the variables on the project's NPV or IRR

While performing sensitivity analysis, decision maker computes the project's NPV or IRR for each forecast under three assumptions:

a. Pessimistic,
b. Expected, and
c. Optimistic

Sensitivity Analysis: Illustration

The financial manager of a food processing company is considering the installation of a plant cost Rs. 1 crore to increase its processing capacity. The expected values of the underlying variables are given as follow. Life of project is 7 yrs.

Sr. No.	Particulars	Amount/units
1	Investment (Rs.'000)	10000
2	Sales volume (units '000)	1000
3	Unit selling price (Rs.)	15
4	Unit variable cost (Rs.)	6.75
5	Annual fixed costs (Rs.'000)	4000
6	Depreciation (%) (WDM)	25%
7	Corporate tax rate (%)	35%
8	Discount rate (%)	12%

Sensitivity Analysis: Illustration

Calculation of Net Cash Flows

	Cash flows (Rs. '000)							
Year/Particulars	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Investment	-10000							
Revenue		15000	15000	15000	15000	15000	15000	15000
Variable Cost		6750	6750	6750	6750	6750	6750	6750
Fixed Costs		4000	4000	4000	4000	4000	4000	4000
Depreciation		2500	1875	1406	1055	791	593	445
Pre-tax profit		1750	2375	2844	3195	3459	3657	3805
Taxes @35%		613	831	995	1118	1211	1280	1332
Profit After Taxes		1138	1544	1849	2077	2248	2377	2473
Net Cash flow	-10000	3638	3419	3255	3132	3039	2970	2918

The project's NPV at 12% is
NPV = 4829 >0
Accept the project

Sensitivity Analysis: Illustration

Forecast under different assumptions

	Variables	Pessimistic	Expected	Opptimistic	% Change
1	Sales Volume (unit'000)	750	1000	1250	25%
2	Units selling price	12.75	15	16.5	15%
3	Units variable cost	7.425	6.75	6.075	10%
4	Annual fixed cost (Rs.'000)	4800	4000	3200	20%

Now, project's NPV are recalculated as below

	Variables	Pessimistic	Expected	Optimistic
1	Sales Volume (unit'000)	-1289	4829	10948
2	Units selling price	-1845	4829	9279
3	Units variable cost	2827	4829	6832
4	Annual fixed cost (Rs.'000)	2456	4829	7203

Sensitivity Analysis: Illustration

- The above NPV calculation under different assumption shows project's sensitivity
- The project does not seem to be that attractive with change in assumptions.
- The most critical variable is sales volume, followed by the units selling price.
- If volume decline by 25%, NPV of the project becomes negative. Similarly, if the unit selling price falls by 15%, NPV is negative.

Evaluation of Sensitivity Analysis

Merits:

- It shows how robust or vulnerable a project is to changes in the values of the underlying variables
- It indicates where further work may be done. If the NPV is highly sensitive to changes in some factor, it may be worthwhile to explore how the variability of the that critical factor may be contained.
- It is intuitively very appealing as it articulates the concerns that project evaluators normally have.

Demerits:

- It merely shows what happens to NPV when there is a change in some variables, without providing any idea of how likely that change will be
- Typically, in sensitivity analysis only on variable is changed at a time. in the real world, however, variables tend to move together.
- It is inherently a very subjective analysis. The same sensitivity analysis may lead one decision maker to accept the project while another may reject it.

Scenario Analysis

- Variables are interrelated.
- Therefore, look at the plausible scenarios, representing a consistent combination of variables.

- For eg. a project may be evaluated under three different scenarios:
  1. The base case scenario where the demand and price are expected to be normal
  2. The scenario where the demand is high, but the price is low.
  3. The scenario where the demand is low, but the price high.

Scenario Analysis

Firm often do a different kind of scenario analysis in which the following scenarios are considered.

- **Optimistic scenario:** High demand, high selling price, low variable cost, and so on.
- **Normal scenario:** Average demand, average selling price, average variable cost, and so on.
- **Pessimistic scenario:** Low demand, low selling price, high variable cost, and so on.

➢ *Scenario analysis, if extended, becomes* ***Simulation Analysis.***

Scenario Analysis: Illustration

In our previous example, it may be possible to

- Increase sales volume to 12,50,000 units (25% increase) if the company reduces unit selling price to Rs. 13.50 (10 % reduction), resorts to aggressive advertisement campaign, thereby increasing unit variable cost to Rs.7.10 (5 % increase) and fixed cost to Rs. 44,00,000 (10 percent increase).
- This scenario generates positive NPV as shown below

	Variables	Expected	Scenario Assumptions
1	Sales Volume (unit'000)	1000	1250
2	Units selling price	15	13.5
3	Units variable cost	6.75	7.1
4	Annual fixed cost (Rs.'000)	4000	4400

Scenario Analysis: Illustration

NPV Calculations at 12% for Scenario:

		Cash flows (Rs. '000)							
	Year/Particulars	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1	Investment	-10000							
2	Revenue		16875	16875	16875	16875	16875	16875	16875
3	Variable Cost		8875	8875	8875	8875	8875	8875	8875
4	Fixed Costs		4400	4400	4400	4400	4400	4400	4400
5	Depreciation		2500	1875	1406	1055	791	593	445
6	Pre-tax profit		1100	1725	2194	2545	2809	3007	3155
7	Taxes @35%		385	604	768	891	983	1052	1104
8	Profit After Taxes		715	1121	1426	1654	1826	1955	2051
9	Net Cash flow	-10000	3215	2996	2832	2709	2617	2548	2496
	PVIF @ 12%		0.893	0.797	0.712	0.636	0.567	0.507	0.452
	PV of Cash Flow		2871	2388	2016	1723	1484	1292	1128
	Total of PV of Cash Flow		12902						
	Initial Cash Outlay (Co)		-10000						
	NPV		**2902**	**>0**					

Break-Even Analysis

- How much should be produced and sold at a minimum to ensure that the project does not lose money.
- This is called break-even analysis and the minimum quantity at which loss is avoided is called the break-even point.
- Most popular way is financial break-even analysis
- Capital budgeting decisions involve uncertainty about:
  - Sales volume
  - Price realization
  - Costs
  - Demand for new products or technologies
- Break-even analysis helps by:
  - Quantifying the margin of safety
  - Showing the sensitivity of profits to changes in volume
  - Indicating how vulnerable a project is to demand shortfalls

Illustration

A company plans to invest in an AI-based SaaS platform.

Fixed Costs (servers, AI models, salaries): Rs. 60,00,000 pa

Selling price per subscription: Rs.10,000

Variable cost per subscription: Rs. 4,000

Contribution per unit = 10000-4000 = Rs.6000

BEP (units) = 6000000/6000 = 1000 units or subcriptions

- If expected sales = 2,500 subscriptions, margin of safety is high that means **low risk**
- If expected sales = 1,100 subscriptions, margin of safety is thin that means **high risk**

Decision Tree Analysis

- Employ to take the sequential investment decisions

Decision tree analysis helps manage risk by:

- Explicitly incorporating probabilities
- Capturing downside and upside risk
- Valuing managerial options (expand, delay, abandon)

Steps in Decision Tree Analysis

1. Identify decision points (invest, expand, abandon)
2. Identify chance events and possible outcomes
3. Assign probabilities to outcomes
4. Estimate cash flows for each path
5. Discount cash flows to present value
6. Compute Expected Monetary Value (EMV)
7. Choose the alternative with the highest EMV

Illustration

- A company is considering launching an AI-driven analytics platform.

Initial Investment (Year 0): Rs. 50 lakhs

After one year, demand may be:

High demand

- Probability = 0.6; NPV = Rs. 40 Lakhs

Low demand

Probability = 0.4; NPV = Rs. -10 Lakhs

THANK YOU!!!